

Colby Dobson

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EDUCATION

University of San Francisco, San Francisco, CA

- President, Club Men's Volleyball August 2023 - May 2026
- B.S. in Computer Science August 2021 - May 2025
 - *Minor in Mathematics*
- M.S. in Computer Science; AI Concentration August 2023 - May 2026

COURSEWORK

Operating Systems, Systems Foundations, Advanced Algorithms, Compilers, Software Development Lifecycle, Cloud Computing, Programming Language Paradigms, Software Development, DevOps, Bayesian Deep Learning, Machine Learning, Programming Synthesis, Advanced Linear Algebra, Linear Regression, Full Stack Web-Dev

PROJECTS

Embedded Systems Arcade Game

Spring 2025 - Present

- Architecting and developing a distributed embedded arcade system across Raspberry Pis with custom hardware interfaces, real-time input arbitration, and synchronized communication
- Implemented fault-tolerant low-level communication, GPIO/UART signaling, and state-coordination protocols for reliable timing-sensitive multi-device interaction under I/O and resource constraints

RISC-V Emulator

Spring 2023

- Emulated a RISC-V processor with C to execute binary instructions/programs.
- Designed digital circuits to build a processor using computer architecture and hardware principles.

Reverse Proxy

Fall 2024

- Led a team of engineers in partnership with Consilient Labs to develop a secure, scalable, and efficient reverse proxy as the primary layer of security for a production API
- Coordinated with the client to align system design with their needs, prioritizing security through fine-grained permission controls and robust access management

Alpha-Zero Style Go Engine

Spring 2026

- Built a Go engine from scratch, progressing to an AlphaZero-style self-play system with policy/value CNNs, MCTS, champion-gated training, and live model evaluation tooling

Agentic Assistant

Spring 2026 - Present

- Working cross-functionally to build a distributed, multi-user AI assistant platform with highly available backend services and agent-based task execution
- Helped design service architecture for orchestration and memory persistence

TECHNICAL SKILLS

Programming Languages: Rust, C++, C, Python, Java, TypeScript, JavaScript, Assembly (RISC-V)

Systems: Linux, Embedded Systems, UART, TDMA, GPIO, Concurrency, Multithreading, Hardware/Software Integration, Computer Architecture, Digital Logic, Protocol Design, Distributed Systems

Infrastructure: Git, Docker, AWS, CI/CD, Microservices, Networking, Actions, K8s, Testing, REST, SQL, noSQL, OAuth,

AI/ML: CNN, Transformer, PyTorch, Regression, LLM Integration, Agent Workflows & Infrastructure, RAG, Tools, Skills

WORK EXPERIENCE

Research Assistant

June 2026 - Present

- Developed and evaluated a foundation model for robust redshift prediction across astronomical surveys
- Designed controlled experiments, cross-survey benchmarks, and scalable HPC training pipelines

University of San Francisco, Computer Science Tutor & TA

August 2024 - December 2025

- Quickly debug and identify issues while explaining complex topics at an elementary level

Sláinte Volleyball Club, Head Coach

September 2021 - July 2025

- Coordinated practices and tournaments, fostered development, and maintained a good environment